

Clinical Case Study #1:

Adaptive Decision-Making in Head & Neck Definitive Chemoradiation

Description

A 58-year-old patient with locally advanced oropharyngeal squamous cell carcinoma is receiving definitive chemoradiation to the primary tumor and bilateral neck.

At simulation, the patient was immobilized with a five-point mask, and the original plan met target coverage and OAR goals. By week 4, however, the patient has lost 12% of body weight. The primary tumor and nodal disease have visibly regressed, the mask has loosened, and repeat daily imaging demonstrates reduced neck separation, shifting parotid position, and increasing concern that the high-dose region is drifting closer to the spinal cord. The treatment team notes an increasing hotspot and loss of intended parotid sparing. While the therapists are still able to achieve daily setup, the team is no longer confident that setup correction alone can preserve the intended dosimetric balance.

Your group must decide whether this patient should undergo adaptive replanning, whether adaptation should be triggered by anatomic change, dosimetric change, or both, and how to balance potential dosimetric benefit against workflow, staffing, and resource considerations.

Discussion Questions

- What anatomic, setup, or dosimetric changes would be sufficient to trigger adaptation in this case?
- Which targets and OARs should be reassessed most carefully?
- What should be the main adaptive goal: preserving target coverage, reducing toxicity risks, maintaining safety margins, or some combination?
- Would you favor IGRT alone, offline adaptive replanning, or online adaptation?
- How would workflow, staffing, treatment planning capacity, QA requirements, and institutional resources influence your recommendation?
- If resources are limited, which patients should be prioritized for ART?

Learning Goals

By the end of this case discussion, participants should be able to:

- Identify realistic anatomic, setup, and dosimetric triggers for adaptive radiation therapy in head and neck cancer.
- Apply adaptive decision-making to a patient with evolving anatomy, tumor regression, weight loss, and changing OAR relationships.
- Balance target coverage, toxicity reduction, safety, and implementation burden when developing a practical ART recommendation.
- Compare the advantages and limitations of continued IGRT, offline replanning, and online adaptation in a real-world clinical scenario.

Clinical Case Study #2:

MR-Guided Online Adaptive Mesenteric Node SBRT

Description

A 53-year-old patient with oligometastatic colorectal cancer has a single mesenteric lymph node that is recommended for SBRT. During simulation, the lymph node shifts more than 1cm between the CT and MRI simulation and is in close proximity to multiple loops of small bowel.

Planning goals are met for 50 Gy in 5 fractions while limiting the adjoining small bowel to 0.5 cc to 33 Gy. The patient undergoes online plan adaptation on day 1, but immediately prior to the beam turning on, confirmation imaging shows the lymph node has moved again. The adaptive process has already required a long decision time, and the patient has become increasingly uncomfortable and verbally aggressive to the therapists. The team is concerned that extending the adaptive session may further increase the risk of intrafraction motion.

Your team must decide whether to proceed with the current treatment, repeat or revise the adaptive workflow, defer treatment, or consider an alternative management approach, while balancing and prioritizing different aspects of the ART workflow.

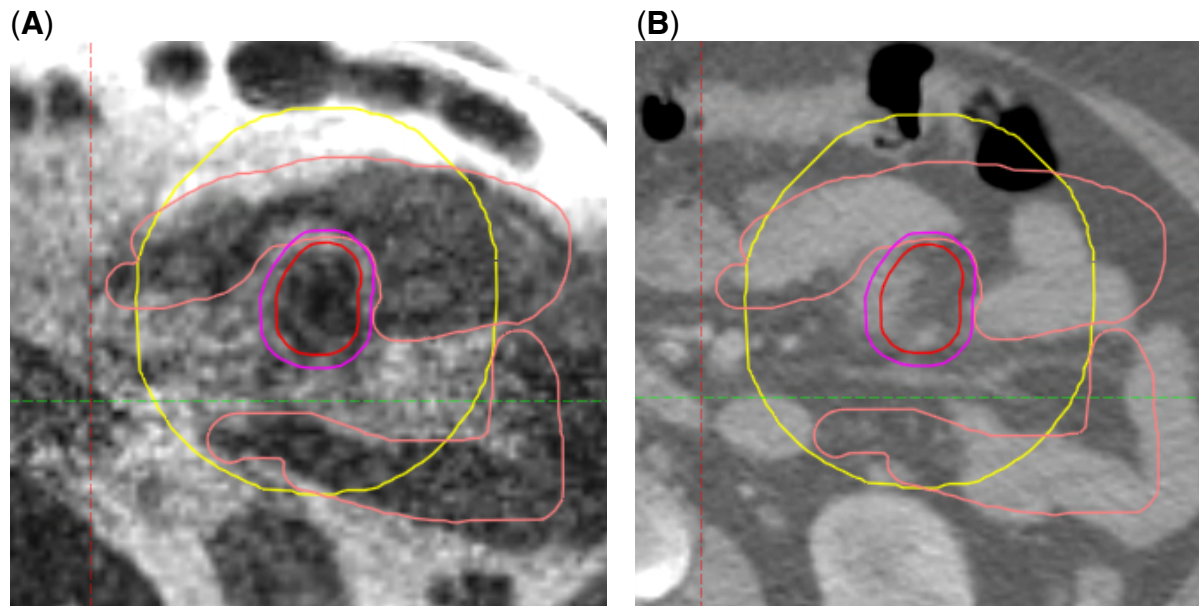


Figure: (A) Simulation MRI and (B) Simulation CT: Target = Red, PTV = Maroon, 3cm Ring = Yellow, and Small Bowel = Peach.

Discussion Questions

- What clinical, anatomic, dosimetric, and workflow factors should drive the decision to adapt on a given treatment day?
- How should the team balance target coverage against OAR sparing in a high-dose SBRT setting?
- How should the team manage a target with both interfraction and intrafraction motion, particularly when motion appears stochastic or unpredictable?
- When does online adaptation provide meaningful benefit, and when might the adaptive process itself introduce additional risk?
- How should treatment efficiency, patient comfort, behavioral concerns, and motion management influence the treatment decision?
- What roles should different team members play in making a safe and practical adaptive treatment decision?
- Under what circumstances should the team proceed, pause, re-image, re-adapt, re-simulate, change immobilization or motion management strategy, or defer treatment?

Learning Goals

By the end of this case discussion, participants should be able to:

- Apply online adaptive radiotherapy principles to a realistic high-risk GI SBRT scenario.
- Evaluate tradeoffs among plan quality, treatment time, patient tolerance, and motion risk.
- Discuss when online adaptation provides meaningful clinical value.
- Recognize the importance of multidisciplinary decision-making in online ART workflows.

Clinical Case Study #3:

Lung ART with Atelectasis Resolution and Motion Uncertainties

Description

A 67-year-old patient with locally advanced lung cancer is receiving definitive radiotherapy for a centrally located tumor near several critical structures. At simulation, the patient has partial atelectasis, and the original treatment plan was generated using 4DCT with an ITV-based approach. The target is in close proximity to the esophagus, proximal bronchial tree, heart, and spinal canal, making the plan highly sensitive to anatomic and motion-related changes.

At approximately fraction 15 of 30, daily CBCT and a repeat verification scan demonstrate partial re-expansion of the previously collapsed lung. This has resulted in a noticeable shift of the primary tumor and involved nodal volumes relative to the esophagus, proximal bronchial tree, and heart. The restoration of lung volume has altered the spatial relationship between the target and nearby OARs compared with the original planning anatomy. Additionally, the treatment therapists note that the patient's breathing pattern has appeared different from earlier fractions, raising concern that the original motion envelope and ITV may no longer accurately represent the current extent or direction of respiratory motion.

Your team must determine whether the observed changes justify offline adaptive replanning, whether repeat motion assessment is needed before modifying margins, and whether the potential dosimetric benefit of adaptation outweighs the uncertainties introduced by tumor regression, changing lung density, and evolving respiratory motion.

Discussion Questions

- Which observed change is most clinically significant in this case: resolution of atelectasis, tumor shift, pleural or lung volume change, or motion uncertainty?
- How should changes in respiratory motion affect the decision to adapt or replan?
- What should be the primary goal of adaptation in this scenario: improving target coverage, reducing dose to nearby OARs, reassessing margins and motion, or some combination?
- What are the risks of continuing the original plan without intervention?

Learning Goals

By the end of this case discussion, participants should be able to:

- Integrate anatomic change and motion into ART decision-making.
- Distinguish when anatomic change alone may justify replanning.
- Evaluate the interaction between adaptive replanning and motion management in thoracic radiotherapy.
- Weigh the benefits of adaptation against workflow complexity and uncertainty in evolving thoracic anatomy.

Clinical Case Study #4:

Re-Irradiation in the Gray Zone

Description

An 82-year-old patient previously received 70 Gy in 35 fractions with concurrent chemotherapy for oropharyngeal cancer 3 years ago at an outside institution. The patient now presents with an ipsilateral nodal recurrence in the left upper neck, adjacent to the prior high-dose region, close to the spinal cord, carotid artery, mandible, and pharyngeal constrictors. Salvage surgery is not feasible.

Available outside records are incomplete. The current team has a treatment summary documenting the prior prescription dose and general field arrangement, but the original images, plan, dose, and structure sets are unavailable. Current imaging shows post-treatment fibrosis and distorted anatomy, making accurate estimation of prior dose to nearby OARs difficult. The team believes re-irradiation may offer meaningful local control or symptom prevention. However, there is substantial uncertainty about cumulative dose to nearby OARs and whether prior dose can be reconstructed with enough confidence to proceed safely.

Your team must decide whether re-irradiation is appropriate, what prior-treatment information is required, whether adaptive radiotherapy could improve safety or confidence, and what additional QA, peer review, and multidisciplinary review are needed before proceeding.

Discussion Questions

- What minimum prior-treatment information is needed before considering re-irradiation?
- How should the prior dose be estimated or reconstructed when historical treatment data are incomplete?
- What level of uncertainty in cumulative dose is acceptable, and how should that uncertainty be communicated to the patient and care team?
- Could ART meaningfully improve safety, confidence, or plan robustness in this case, or is the dominant challenge uncertainty from the prior treatment course?
- What additional QA, peer review, risk assessment, or multidisciplinary review steps are needed before proceeding?

Learning Goals

By the end of this case discussion, participants should be able to:

- Examine how cumulative dose uncertainty affects re-irradiation decision-making.
- Discuss the roles and limitations of dose accumulation, image registration, prior dose reconstruction, QA, and safety review.
- Evaluate whether ART can add value in a high-risk re-treatment setting.
- Integrate safety, risk communication, patient counseling, and multidisciplinary review into adaptive decision-making.
- Recognize when technical uncertainty may limit the safe or meaningful use of ART.

Clinical Case Study #5:

Robust Optimization Vs. Adaptive Replanning in Proton Therapy

Description

A 12-year-old patient with a locally advanced sinonasal/skull-base tumor is being treated with proton therapy to spare the optic nerves, optic chiasm, brainstem, and temporal lobes, while reducing integral dose to developing normal tissues.

The initial treatment plan was robustly optimized for setup and range uncertainties, and treatment began normally. During week 2, however, on-treatment imaging shows partial tumor shrinkage, increased sinus aeration, and modest weight loss. The physician anticipates continual changes over the course of treatment and asks whether the original robust plan is still adequate. The physicist is concerned that the anatomic changes may have altered proton range, potentially shifting high-dose regions toward OARs, including the optic apparatus and brainstem.

Your group must determine whether to continue with the current robustly optimized plan, perform offline adaptive replanning, consider online adaptation if available, or pursue an alternative management strategy. An additional question is whether a photon adaptive backup plan or hybrid proton-photon approach may be more practical than repeated adaptive proton replanning.

Discussion Questions

- What imaging or dosimetric findings would justify adaptive replanning despite robust optimization? Which anatomic changes are most important for evaluating proton range and dose delivery?
- What strategy would you recommend: continue the current plan, perform offline replanning, consider online adaptation, or use an alternative approach?
- Which imaging modality would be sufficient for reliable proton adaptation: repeat CT, in-room CT, or CBCT?
- How should pediatric priorities influence the adaptive threshold?
- How should the team balance potential adaptive benefit against workflow burden and treatment delay?
- When might a photon adaptive backup plan or hybrid proton-photon approach be more practical than repeated proton adaptive replanning?

Learning Goals

By the end of this case discussion, participants should be able to:

- Distinguish robust optimization from adaptive replanning and recognize why adapted proton plans still require robustness evaluation.
- Recognize how beam-path changes (such as air-tissue interface shifts, weight loss, and tumor regression) can affect proton dose delivery.
- Evaluate when the clinical value of proton ART justifies the added operational complexity, and when a different strategy may be more practical.
- Compare proton adaptation with alternative strategies, such as photon adaptive backup planning or hybrid proton-photon treatment.

Clinical Case Study #6:

PET-Guided ART: Research Signal or Biologically Guided Adaptation?

Description

A 48-year-old patient with node-positive head-and-neck cancer is undergoing definitive chemoradiation and has relatively stable anatomy on daily setup imaging. During treatment, an interim research PET scan shows a heterogeneous metabolic response. The primary tumor demonstrates decreased uptake, while one nodal region shows persistent avidity.

The physician and clinical team are considering whether this PET information should influence the adaptive radiotherapy strategy, including the possibility of a biologically guided dose boost to the persistently avid nodal region.

Your group is asked to evaluate whether the imaging findings are sufficiently robust, reliable, and clinically actionable to support adaptation, or whether the treatment approach should remain anatomy-based within the current standard workflow.

Guiding Questions

- What image quality limitations (such as signal-to-noise ratio, spatial resolution, motion, reconstruction method, or reconstruction bias) could affect interpretation of the PET response?
- How reliable are relative activity changes in the intra-treatment setting, particularly in the setting of image noise, differences in uptake time, scanner variability, patient positioning, and treatment-related inflammation?
- Should biologic or metabolic response ever trigger adaptation in routine clinical care, or should it remain investigational?
- What level of evidence would be required before changing dose based on PET response?
- How would you manage uncertainty in PET-based target definition, including partial-volume effects, segmentation variability, and registration uncertainty?
- What are the potential failure modes, safety concerns, or QA requirements for PET-guided ART?
- Would this scenario be best managed within a prospective clinical trial or protocol-driven workflow?

Learning Goals

By the end of this case discussion, participants should be able to:

- Describe key image quality and quantitative limitations of PET in ART settings.
- Evaluate the promise, limitations, and risks of biologically guided ART.
- Distinguish investigational adaptation from routine clinical implementation.
- Assess how uncertainty in deformable image registration, image quality, target definition, and dose delivery affects the safety and feasibility of PET-guided ART.
- Define practical evidence and QA thresholds that would be needed before implementing biologically guided dose adaptation in clinical practice.